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The British Tertiary igneous province: palaeomagnetism of the dyke swarm along the Sleat coast of Skye

R. L. Wilson *Sub-department of Geophysics, Liverpool University, Liverpool L69 3BX, England*

J. M. Hall *Department of Geology, Dalhousie University, Halifax, Nova Scotia B3H 3J5, Canada*

P. Dagley *Sub-department of Geophysics, Liverpool University, Liverpool L69 3BX, England*

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Summary. A palaeomagnetic investigation has been made of a swarm of more than 400 dykes along the south coast of Skye, Scotland, by the Sound of Sleat.

Seven red lamprophyre dykes have palaeomagnetic directions inconsistent with Tertiary age, and not inconsistent with their previously held Caledonian age. The remaining 409 dykes have palaeomagnetic directions that are consistent with a Lower Tertiary age. We present evidence suggesting that the Tertiary dykes might have been emplaced during a short time, over which the geomagnetic polarity occupied as few as three polarity intervals (NRN or RNR).

Certain 'intermediate' directions of magnetization have also been found, and are presented here.

1 Introduction

For more than a decade, the Liverpool Geophysics Sub-department has been investigating palaeomagnetically and geochronologically several thousand lavas, dykes, sills and associated baked rocks in the British Tertiary igneous province. This paper is a report on the Skye dyke swarm along the coast bordering the South of Sleat. These dykes trend perpendicular to the coast and are intruded into Lewisian gneiss and Torridonian sandstone and grits, except for the westernmost few dykes which are found intruded into Moine schists. There is little doubt that the basic dykes which form most of this swarm are of Tertiary age although no radiometric dates exist on this particularly swarm. However, the swarm includes seven red lamprophyre dykes, visually very distinct from the rest, which have previously been assigned a Caledonian age (Richey 1939).

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Our ultimate objective is to use the accumulated detailed British Tertiary igneous province studies to thrown light on the origin and geological significance of the whole igneous province; and also to study the behaviour of the ancient geomagnetic field. In this process, many smaller individual problems arise (and are sometimes solved); the age of the red lamprophyre dykes being an example of such a problem.

2 Sampling, measurement and data treatment

Our collection extended around the coast of Skye from 156799 to 180825 on the British National Grid. The mean site latitude and longitude are 57.1° N and 354.1° E. We sampled every coastal dyke (over 400) that could be found, down nearly to the low tide level. Dykes were numbered according to J. M. Speight's (1972) previous numbering of this swarm. Additional dykes which we later discovered (often near low tide levels) were inserted into the sequence with additional letters (e.g. 66A). These additional letters have no further significance.

At least four oriented samples were collected from each dyke, using Sun-sighting as a basis for orientation. No sample was taken closer than one dyke width from any adjacent dyke, to avoid serious thermal remagnetization. Our experience suggests this to be a very conservative distance (Ade-Hall *et al.* 1972). Each sample was measured using a parastatic magnetometer after alternating field demagnetization in fields regularly increased in 5 mT steps. Samples which did not display clear directional stability by 25 mT were further demagnetized either to 150 mT in various steps, or until no further stable remanence could be detected.

Dyke-mean final directions of magnetization were calculated using the method of Dagley & Ade-Hall (1970). Each Tertiary dyke mean will be made available to any reader on request to the first author. Several dykes were completely rejected because their directions of magnetization were completely inconsistent.

3 The seven red lamprophyre dykes

Palaeomagnetic data for these lamprophyres are as follow:

Table 1. Data from the red lamprophyre dykes. Dyke codes which are followed by a letter (A, B, . . .) have no significance other than the coding method mentioned in the section in sampling. This mean direction is compared with that for three accepted Devonian reversed directions from the literature.

Dyke code No.	Declination (°)	Inclination (°)	N	α_{95} (°)
AOH307A	192.8	51.2	4	1.9
303A	16.6	80.0	3	5.3
303B		No consistent result		–
304A	186.3	70.1	5	9.7
304C	338.0	88.2	3	11.7
304D	207.1	50.7	4	10.4
367A	40.9	83.8	4	7.4
Mean	196.1	76.8	6 dykes	19.2
Creer DII (1963)	215.8	58.0	8	7.9
Embleton (1968)	232	41	2	–
Sallomy & Piper (1973)	221	41	13	11.5

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The specimens from 303B were not magnetically consistent with each other. The remaining six yielded the data in Table 1, where we compare their mean direction of magnetization with the reversed directions of Creer (1963) (the DII group), Sallomy & Piper (1973), and Embleton (1968). (El-Batroukh 1971, Creer (DI) 1963, and Collinson 1980 are considered unrepresentative of Devonian magnetization, see Collinson 1980). Our mean direction lies about 30° from the mean of the three directions quoted above. Our mean is also 46° from the normal Tertiary direction associated with the 409 Tertiary dykes. Given the α_{95} of 19° on our sample of six dykes, then a modest folding of order 15° between Devonian and Lower Tertiary times would make the lamprophyres' magnetic directions consistent with known Devonian results. The tilts marked on the geological map of Skye give little indication of a *general* tilt trend, and indeed that would be of little use since the *times* of tilting would be unknown.

On the other hand, no tilting since Lower Tertiary times can change the 46° between lamprophyre and basaltic dykes, since they would have that tilt in common.

We can therefore say that the lamprophyres, deviating more than $2\alpha_{95}$ from the Tertiary dykes, are not likely to be of Tertiary age. Less confidently, we can say that a modest folding of 15° prior to emplacement of the Tertiary igenous province, would make the lamprophyres consistent with Devonian magnetizations. Given Richey's (1939) other arguments for Caledonian emplacement, it seems that our magnetizations do not seriously conflict with that view.

4 The 409 Tertiary dykes

We should note that tilt corrections were not possible in the case of these Tertiary dykes intruded into much older rocks. The overall mean palaeomagnetic pole position (Table 2), not tilt corrected, of this dyke swarm may be taken as 162.2° E, 82.3° N ($dP = 1.9°$, $dM = 2.3°$). This pole position is about 10° closer to the present geographic pole than that of other tilt-corrected British Tertiary *lava* piles, including the Skye lavas (see Hall, Wilson & Dagley 1977 and Wilson, Dagley & Ade-Hall 1972), but on the other hand the palaeomagnetic poles of the present swarm agree well with those of the Vaternish *dyke* swarm of Northern Skye (Wilson *et al.* 1974). The significance of all these poles can only be discussed in an overall British Tertiary igneous province paper. Table 2 gives means and statistics for this dyke swarm for various levels of data rejections based on α_{95} for the dyke-mean directions of magnetization. Fig. 1 displays poles for which α_{95} is less than 5.0°.

For the purposes of Table 2, normal and reversed poles have been defined simply as lying north or south of the Earth's equator. From Table 2 we see that rejection of dykes displaying various degrees of scatter alters the mean pole position very little; less than the (dP, dM) errors would suggest. The N and R mean poles are antiparallel to within 0.7°; and yet their axis differs significantly (by 7.7°) from the present geographic pole. This is a classic proof that the magnetization of these dykes is primary, although the proof is weakened somewhat by the small angle (7.7°) involved.

Table 2.

Rejection level	No. of dykes			Pole latitude			Pole longitude			dP, dM		
	N	R	All	N	R	All	N	R	All	N	R	All
No rejections	81	328	409	84.0	82.2	82.5	165.8	155.6	158.0	4.9, 5.8	2.3, 2.8	2.1, 2.5
$\alpha_{95} < 10°$	75	309	384	84.4	82.4	82.8	151.1	159.9	158.2	4.5, 5.3	2.1, 2.6	1.9, 2.3
$\alpha_{95} < 7.5°$	70	300	370	83.5	82.2	82.5	164.1	160.6	161.2	4.0, 4.8	2.1, 2.6	1.9, 2.3
$\alpha_{95} < 5.0°$	60	268	328	82.5	82.8	82.3	165.5	160.8	162.2	3.8, 4.6	2.2, 2.6	1.9, 2.3

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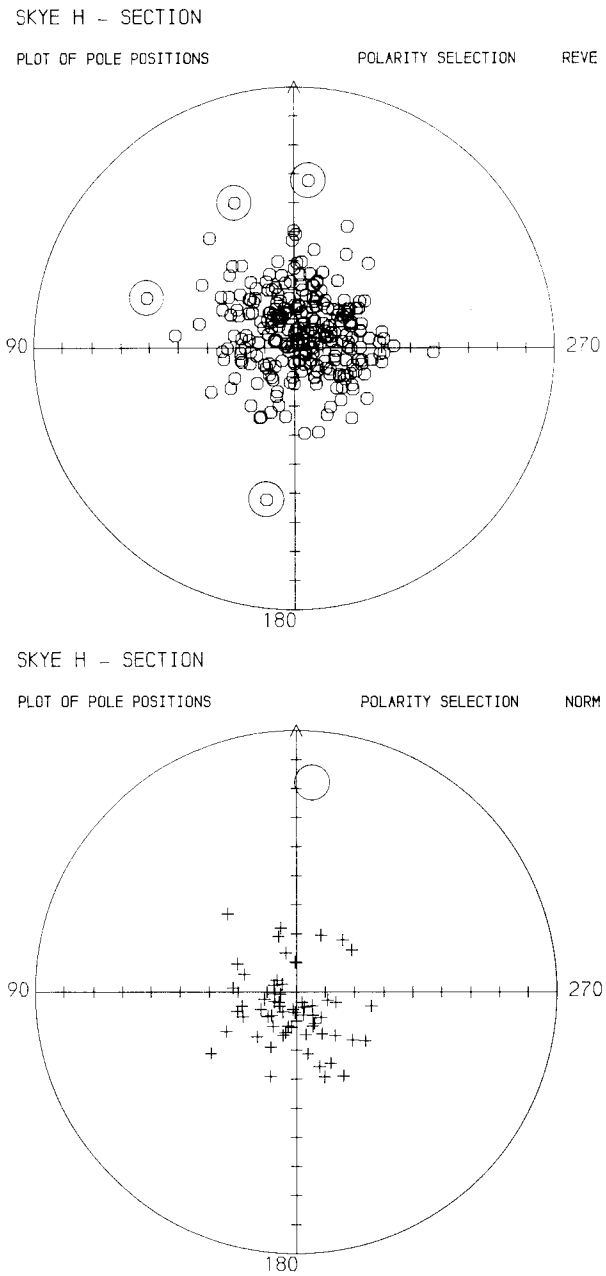


Figure 1. Virtual geomagnetic pole positions for dykes yielding α_{95} errors on mean direction of magnetization less than 5°. Circles enclose five dykes exhibiting clearly intermediate pole positions. The empty circle on the normal plot concerns an intermediate pole for a dyke (H344) with $\alpha_{95}=8.1°$, but which we judge to be a significant intermediate pole position.

The overall mean pole is also far-sided and right-handed (Wilson & McElhinny 1974), but the age of the province is great enough for plate movement to be at least partly and possibly completely responsible for this effect, since a drift rate of only 1.5 cm yr^{-1} could have effected the present polar displacement.

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Table 3. Five unambiguously intermediate dyke magnetizations.

Dyke code No.	N	D (°)	I (°)	α_{95} (°)	VGP latitude	VGP longitude
A03H020	4	120.9	−41.1	2.8	−36.3	71.3
H031	3	10.5	−79.5	1.8	−36.8	169.4
H061	4	157.3	−11.9	0.6	−36.1	23.1
H179	4	179.1	1.7	1.7	−32.1	355.0
H344	4	163.3	67.2	8.1	17.8	5.0

4.1 INTERMEDIATE DIRECTIONS OF MAGNETIZATION

We may define an intermediate direction of magnetization in a way suggested by experimental evidence (Wilson, Dagley & McCormack 1972); it is a direction such that the corresponding virtual geomagnetic pole is farther than 40° from the geographic pole. Accepting for the moment that here we mean the *present* geographic pole, then there are 16 Skye dykes in our collection which are technically intermediate (H006, 018, 020, 026, 031, 034, 051, 052, 061, 121E, 121F, 179, 165A, 300, 344, 364). Many of these are barely intermediate on our definition, and when errors are taken into account, only five of these dykes remain completely and clearly intermediate. If, on the other hand, we consider that the geographical pole should be the *ancient* pole at the time of dyke intrusion, then even fewer than 16 intermediates will exist, but the above five still remain clearly intermediate. They are plotted as heavily circled symbols in Fig. 1 and are listed in Table 3.

These intermediate directions of the ancient geomagnetic field must be considered along with all other intermediates in a further paper involving the entire British Tertiary igneous province.

5 Dyke cutting relationships, and the geomagnetic polarity history during creation of the British Tertiary igneous province

There are 34 cases in this Skye collection in which one dyke cuts or touches another or others. These dykes were all sampled far enough from their contacts to avoid heating effects on their magnetic remanences (more than one dyke width – see Ade-Hall *et al.* 1972). We have deduced the relative ages of the contacting dykes by observing either chilling or cutting relationships. In all but three of these cases, the polarities of the earlier and the later dykes agree. The average angle between the magnetizations of any two such contacting dykes is 10.6°, which seems compatible with what would be expected on the basis of normal secular variation plus measurement errors. The three cases involving distinct polarity differences are listed in Table 4.

Table 4. Data from contacting dykes, relevant to the geomagnetic inversion history during formation of the British Tertiary igneous province.

Dyke code No.	Age relation	D (°)	I (°)	α_{95} (°)	Polarity	Indicated polarity sequence
H052	Oldest	280.7	−66.9	8.7	R	
H051	Both younger	147.8	−29.6	3.4	R	R→N
H053	than H052	295.9	63.5	14.2	N	
H103	Older	355.1	66.7	3.7	N	N→R
H104	Younger	246.4	−77.7	4.1	R	
H216	Older	13.0	64.8	5.2	N	N→R
H217	Younger	226.3	−63.4	4.5	R	

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H053 has not a very well-defined direction, but all four samples clearly remained normal during all of their demagnetization steps. The simplest time sequence of geomagnetic polarity compatible with Table 4 is either NRN or RNR. Furthermore, the fact that *few* (three out of 34) contacting dykes have opposite polarities suggests that few inversions took place during the insertion of the whole dyke swarm; otherwise one would have expected about half of the 34 dyke pairs to have contrasting polarities and half of the dyke pairs to have similar polarities. There is little evidence in the British Tertiary igneous province of any necessity for a more complicated history than three successive polarity periods (Dagley & Hodgson 1979; Mussett, Dagley & Skelhorn 1980). Indeed, Dagley & Hodgson suggest that the Arran polarity sequence is only compatible with RNR (not NRN). With the exception of studies on the Island of Lundy all of the evidence (Mussett 1980) points to a brief period – perhaps less than 2 or 3 Myr – for the creation of the entire province, taking into account our knowledge (Hailwood *et al.* 1979; La Breque, Kent & Cande 1977) of the frequency of geomagnetic inversions near to 58 Ma.

6 Discussion

The South Skye dyke swarm has provided us, first, with the answers to two special and local problems.

(a) The palaeomagnetic evidence submitted in this paper makes it quite certain that this swarm is part of the British Tertiary igneous province, with the exception of seven red lamprophyre dykes, six of which have magnetizations not inconsistent with Caledonian age.

(b) The geological evidence summarized by Richey (1939) in assigning a Devonian age to these seven red lamprophyres is moderately supported by the palaeomagnetic evidence in this paper. This evidence also suggests that folding of the host rocks, not necessarily exceeding about 15° of tilt, has occurred since Devonian times.

This swarm has also contributed to, secondly, three more general questions.

(a) The behaviour of specimens during AF demagnetization, and the precise palaeomagnetic antipolarity of N and R mean poles, suggest good palaeomagnetic stability and a faithful record of Lower Tertiary magnetic field behaviour in most of the dykes sampled. The deviation of the ancient polar axis from the present rotation axis is in reasonable agreement with other evidence from this Tertiary province. This dyke swarm defines an ancient pole position which is 7.7° from the present geographic pole, with a 95 per cent error of the mean near 2.1° (Table 2).

(b) The cutting and chilling relationships between physically contacting dykes of opposite polarity, and the fact that in only three cases out of 34 do contacting dykes have opposite magnetic polarity, both lead us to suggest that few polarity inversions occurred during the emplacement of the swarm, and that the minimum polarity sequence necessary to accommodate the evidence is either NRN or RNR. Other evidence (Dagley & Hodgson 1979) suggests RNR during the Arran Tertiary activity.

(c) We have five clear-cut cases of intermediate directions of magnetization.

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